

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) II Semester Examination, 2013-14

Botany**Paper: BOB-201 — Microbiology, Plant Pathology, Cytology and Genetics**

Time: Three Hours

Maximum Marks: 70

Note: Attempt five questions in all, selecting two questions each from Section A and B. Question No. 9 (Section C) is compulsory. Marks allotted to each question are given on the right.

SECTION – A

Question 1. With the help of suitable diagrams describe the structure of the bacterial cell wall and flagella. [15]

Question 2. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Plasmids
- (b) Endospore
- (c) Lytic cycle

Question 3. Write critical notes on any two of the following: [7.5 × 2 = 15]

- (a) Late blight of potato
- (b) Loose and covered smut
- (c) Transformation in bacteria

Question 4. Give a brief history of discoveries in the field of microbiology focusing on the contributions made by Louis Pasteur and Robert Koch. [15]

SECTION – B

Question 5. Differentiate between intergenic and intragenic interactions, and describe briefly complementary gene interaction with a suitable example. [15]

Question 6. Describe briefly any two of the following: [7.5 × 2 = 15]

- (a) Prophase I of meiosis
- (b) Chromosomal theory of sex determination in plants
- (c) Dominance and co-dominance

Question 7. Why Mendel selected garden pea for his experiments? Briefly explain Mendel's laws of inheritance. [15]

Question 8. Write short notes on any three of the following: [5 × 3 = 15]

- (a) Spindle apparatus
- (b) Crossing over and linkage maps
- (c) Nuclear pore complex
- (d) Chromatin

SECTION – C

Question 9. Give very brief answers, not more than one sentence each, of the following: [10 × 1 = 10]

- (a) Where does lambda DNA integrate on the chromosome of the host bacterium?
- (b) Wilt of Arhar is caused by _____
- (c) What is the importance of denitrification in nature?
- (d) What is ammonification?
- (e) Replication of the bacterial chromosome.
- (f) What does Hfr mean?
- (g) Differentiate diakinesis from Metaphase I of meiosis.
- (h) Define synthetic theory of evolution.
- (i) Differentiate phenotype from genotype.
- (j) What are lethal genes?

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